

Differential Diagnosis Copilot

Benchmark results for team review · 2026-05-18

Summary. We built two complementary benchmarks on MIMIC-IV (real de-identified hospital data) to test whether a Claude-based diagnostic copilot can surface diagnoses physicians might miss. The headline results are encouraging: **79% recall** on high-stakes diagnoses from 24 hours of workup data, and **94% detection** of algorithmically-defined AKI cases that the treating physicians never recognized. Both numbers are statistically meaningful at our current sample sizes. This document explains what we tested, the main findings, and the gaps that need closing before external claims.

79%

high-stakes recall@15
on n=50 MIMIC cases
at admit+24h

94%

AKI detection rate
on n=100 never-diagnosed
cases at admit

\$0.04

cost per case
with Claude Haiku 4.5
(production-feasible)

What we tested

Two distinct benchmarks, each answering a different question:

- 1. Standard recall benchmark.** Given a patient chart at various timepoints (admission, +24h, +48h, pre-discharge), what fraction of the eventual acute discharge diagnoses does the LLM rank in its top-15? This measures the model's general diagnostic coverage — can it surface what the physician will eventually arrive at?
- 2. Never-detected test (AKI specifically).** We identified 37,095 MIMIC admissions where (a) the patient met KDIGO criteria for AKI by lab values, but (b) AKI was never coded in the discharge ICD list AND (c) AKI was never mentioned in physician notes. These are **genuinely missed diagnoses** — not coding errors. We then test whether the LLM, given only admission data, flags AKI for these patients.

Strategic note. Existing startups (SmarterDx, Iodine) catch diagnoses documented in notes but missed by coders. Our test #2 specifically targets cases where the diagnosis was *never documented* — a fundamentally different problem and value proposition. The 11% of uncoded AKI that was in notes (coding errors) was deliberately excluded from our headline test to keep the comparison clean.

Methodology

Data source

All testing was on **MIMIC-IV v3.1**, the standard public research dataset of ~340,000 de-identified hospital admissions from Beth Israel Deaconess Medical Center (Boston). We use the free-text discharge summaries (MIMIC-IV-Note 2.2), emergency department records (MIMIC-IV-ED 2.2), and the full hospital labs/meds/procedures tables. Access requires PhysioNet credentialing and a Data Use Agreement.

Chart construction (leak-safe)

For each test case we build a sanitized chart at multiple time cutoffs (admission, +24h, +48h, pre-discharge). Critical leak-prevention steps:

- **Discharge summary sections stripped:** Brief Hospital Course, Discharge Diagnosis, Discharge Medications, Discharge Instructions, Follow-up Instructions — anything written with hindsight of the diagnosis.
- **Admission sections retained:** Chief Complaint, History of Present Illness, Past Medical History, Social/Family History, Allergies, Medications on Admission, Physical Exam, ED Triage.
- **Time-filtered events:** labs, microbiology, radiology reports, and prescription orders are filtered to those available by the cutoff time (using each event's *storetime* — when the result became available to the treating physician).

Gold standard

For standard recall benchmark: the patient's actual discharge ICD code list, filtered to exclude chronic comorbidities (hypertension, diabetes, etc.), symptom codes, external-cause codes, and status codes. We score only the **business-relevant acute diagnoses** that the workup was supposed to reveal.

For never-detected AKI test: the algorithmic KDIGO Stage 1+ criterion — serum creatinine rise of ≥ 0.3 mg/dL in any 48-hour window, OR creatinine $\geq 1.5\times$ the prior baseline. This is the standard clinical definition; no physician documentation required.

Model under test

Claude Haiku 4.5 is our primary model. We tested Sonnet 4.6 and Opus 4.7 on the same MTSamples benchmark in an earlier prototype — Sonnet beat Haiku marginally on small consult notes, but Opus was *worse* than Haiku on real MIMIC charts at 17 $\times$ the cost. Haiku is the right production choice for this task.

Scoring

Standard benchmark: for each gold diagnosis, an LLM judge (also Haiku) decides whether any of the candidate diagnoses clinically matches. Recall = fraction of gold diagnoses matched in the top-K candidates.

AKI never-detected test: simple regex match for AKI / acute renal failure / acute kidney injury in the candidate list. No judge needed — the question is binary.

Standard benchmark — what the LLM recovers

n=50 stratified MIMIC-IV admissions, 1 per specialty bucket × 5 (cardiac, GI, neuro, infectious, etc.). 120 high-stakes diagnoses total across all cases. Tested at 4 time cutoffs.

Headline numbers

Cutoff	All-acute r@15	High-stakes r@15	Non-high-stakes r@15
admit (HPI + ED only)	58%	73% (88/120)	51%
+24h (peak)	63%	79% (95/120)	55%
+48h	62%	78% (93/120)	54%
pre-discharge	59%	77% (92/120)	51%

Total run cost: \$1.89 with Haiku. ~33 min wall-time.

Three findings worth highlighting

1. The curve plateaus at +24h. Adding more chart data beyond the first day does not meaningfully improve recall. Most predictive signal is captured by 24 hours of workup. This has product-design implications — the AI's value is concentrated at early time points.

2. The model is biased toward high-stakes conditions. Gap between high-stakes recall (sepsis, AKI, MI, PE, pneumonia, stroke, HF, etc.) and non-high-stakes recall is +22 to +26 percentage points across all cutoffs. The LLM uses its 15 ranked slots on clinically dangerous diagnoses, not incidental findings. This is the right inductive bias for a safety-net copilot.

3. Miss-evidence audit shows the data is sufficient. Of 153 missed gold diagnoses, we asked Sonnet 4.6 to read the actual chart and judge whether evidence was present. Results:

	All misses	High-stakes only
EVIDENCE_SUFFICIENT — model failure	63% (96/153)	64% (18/28)
EVIDENCE_PARTIAL — borderline	29%	36%
EVIDENCE_ABSENT — data limit	8%	0%

Zero high-stakes misses are due to missing data. For dangerous conditions, the chart always has at least partial evidence — the LLM layer is the bottleneck, not the data pipeline.

What we tried that didn't help

Three optimizations were tested and did **not** beat the simple Haiku baseline:

- **Opus 4.7 + same prompt:** 71% pre-discharge HS recall (vs Haiku 79%). 17× cost, worse accuracy.
- **Heavy rule-based prompt with extraction checklists:** 71% pre-discharge HS recall (-8 pp). More verbose output, fewer correct diagnoses.
- Both produced more output tokens for fewer right answers. The bottleneck is not model capability or prompt complexity.

Never-detected test — the bigger story

Designed to test the exact product claim: catching diagnoses physicians genuinely missed (not coding errors). AKI as the wedge condition because of strong literature support (Cammarata 2024: 68% AKI undercoding) and clean algorithmic criteria (KDIGO).

How we found the test cases

Starting from all 320,679 MIMIC admissions with ≥ 2 creatinine values, we applied a multi-step filter:

Stage	Count	% of eligible
Eligible (≥ 2 Cr values)	320,679	100%
Meet KDIGO Stage 1+ criteria	106,980	33%
KDIGO+ AND no AKI in discharge ICD codes (Tier 1)	56,333	18%
KDIGO+ AND no AKI in ICD AND no AKI mentioned anywhere in notes (Tier 2)	37,095	12%
Stratified sample for LLM test	100	—

Our finding: 52.7% AKI undercoding rate (vs Cammarata 2024 literature finding of 68%; differences likely explained by baseline creatinine definition and BIDMC's academic-medical-center coding rigor).

Critically — only 6,272 of the 56,333 uncoded AKI cases actually had AKI mentioned in physician notes (would be addressable by coding-error products). The other 37,095 are genuine diagnostic misses — exactly the cases this product exists to catch.

LLM detection results

We stitched chart inputs for 100 stratified Tier 2 cases (15 severe AKI by ratio $\geq 3\times$, 3 moderate, 82 mild) and asked Haiku for top-15 diagnoses at admission cutoff and +24h cutoff.

Cutoff	hit@5	hit@15
admit (HPI + ED only)	49% (49/100)	94% (94/100)
+24h	32% (32/99)	65% (64/99)

By severity (admit cutoff, hit@15):

- **Severe AKI** (Cr ratio $\geq 3\times$): 93% (14/15)
- **Moderate AKI** (Cr ratio 2– $3\times$): 100% (3/3)
- **Mild AKI** (Cr ratio 1.5– $2\times$, the hardest): 94% (77/82)

Mild AKI is just as detectable as severe. This matters — Stage 1 (mild) AKI is what physicians most commonly miss, and it's where the literature shows the biggest under-recognition gap.

Total cost for the 100-case run: **\$1.56**. ~35 minutes wall-time.

Honest caveats — what we have NOT measured

These results are real and meaningful, but they're not a complete diagnostic-accuracy picture. Five specific gaps that limit external claims:

1. False-positive rate

We measured sensitivity (true positive rate). We have NOT measured specificity. If the LLM flags AKI on a high fraction of patients who don't have it, the 94% sensitivity is meaningless. **Estimated fix: \$2 and 30 minutes** — run the same prompt on 100 KDIGO-negative admissions, count false AKI flags.

2. External validity

MIMIC-IV is one academic medical center (BIDMC), ICU-heavy. Community hospital data, ambulatory data, and non-US health systems may produce different numbers. Requires partnership with a design partner to test. We've not yet established the gap.

3. Clinical impact

We measured whether the LLM flags a diagnosis. We have NOT measured whether physicians would act on the flag, or whether acting on it would change patient outcomes. This requires a prospective interventional study — years away.

4. AKI-only for the headline result

The 94% number is specifically about AKI. We chose AKI because (a) KDIGO criteria give algorithmic ground truth, and (b) it has the strongest literature support. Other conditions (sepsis, PE, stroke) will have different detection rates. We should replicate for at least sepsis before generalizing.

5. Judge variance and synonym noise

Our standard benchmark uses an LLM judge to decide whether a candidate diagnosis matches a gold ICD code. The judge introduces ~5-10% noise. The AKI never-detected test avoids this entirely with regex matching, which is why it's more rigorous.

What we'd do next, in priority order

1. Measure false-positive rate (~\$2, 1 hour). Without this, the 94% sensitivity number is one-sided. This is the single most important next experiment.

2. Replicate the never-detected test for sepsis (~\$5, 4 hours). Sepsis has SIRS/qSOFA criteria similar to KDIGO and similar documented undercoding rates. Demonstrates the method generalizes.

3. Find a physician co-founder or design partner. All subsequent validation requires clinical eyes. Defensible numbers from a benchmark only get you so far — physicians need to review the model's actual outputs to call them "clinically meaningful."

4. Scale to n=500 on the standard benchmark. Tighten confidence intervals. Cost ~\$20.

5. Test on non-MIMIC data. Either de-identified data from a design partner, or synthetic ambulatory data (different problem).

Repro. All code, prompts, and aggregate results in the `ml/prototype_b/` directory of this repository. Per-case patient data is gitignored under MIMIC's DUA. Total spend on all experiments to date: under \$20.